IAS PRELIMS/MAINS

SCIENCE & TECHNOLOGY

A COMPREHENSIVE COVERAGE

MODULE - 37

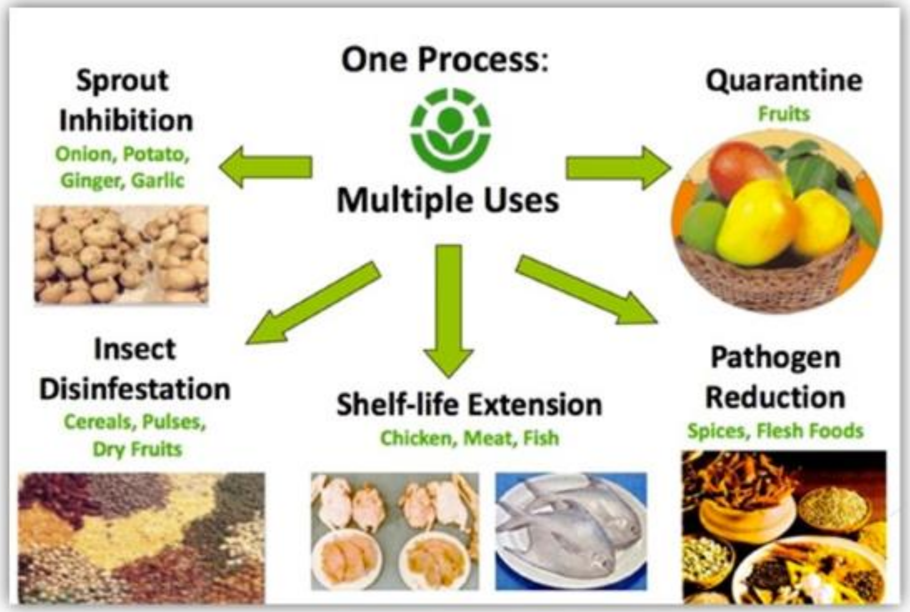
NUCLEAR TECHNOLOGY - 3

NUCLEAR TECHNOLOGY

APPLICATIONS OF NUCLEAR TECHNOLOGY

1 FOOD IRRADIATION

- **Food irradiation (the application of ionizing radiation to food) is a technology that improves the safety and extends the shelf life of foods by reducing or eliminating microorganisms and insects.** Like pasteurizing milk and canning fruits and vegetables, irradiation can make food safer for the consumer.
- **Irradiation does not make foods radioactive, compromise nutritional quality, or noticeably change the taste, texture, or appearance of food.** In fact, any changes made by irradiation are so minimal that it is not easy to tell if a food has been irradiated.



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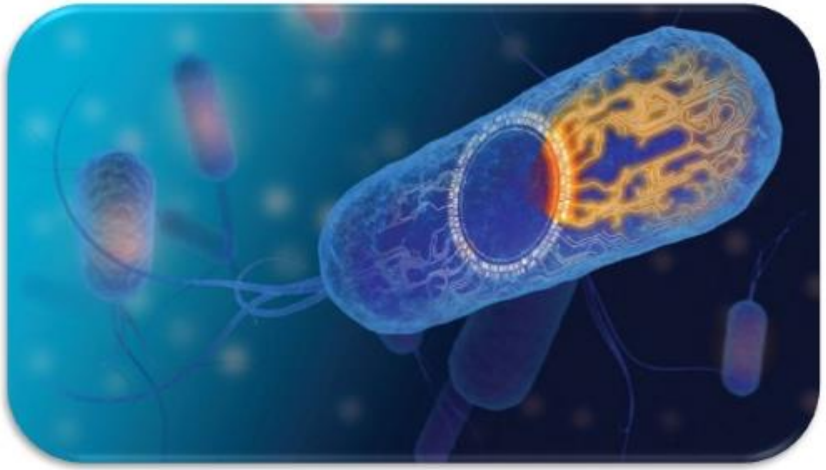
WHY IRRADIATE FOOD?

IRRADIATION CAN SERVE MANY PURPOSES

- **Prevention of Foodborne Illness** – to effectively **eliminate microorganisms** that cause foodborne illness, such as *Salmonella* and *E. coli*).
- **Preservation** – to destroy or inactivate organisms that cause **spoilage and decomposition** and extend the shelf life of foods.
- **Control of Insects** – to destroy insects in or on fruits imported into our country. Irradiation also **decreases the need for other pest-control practices** that may harm the fruit.



WHY IRRADIATE FOOD?

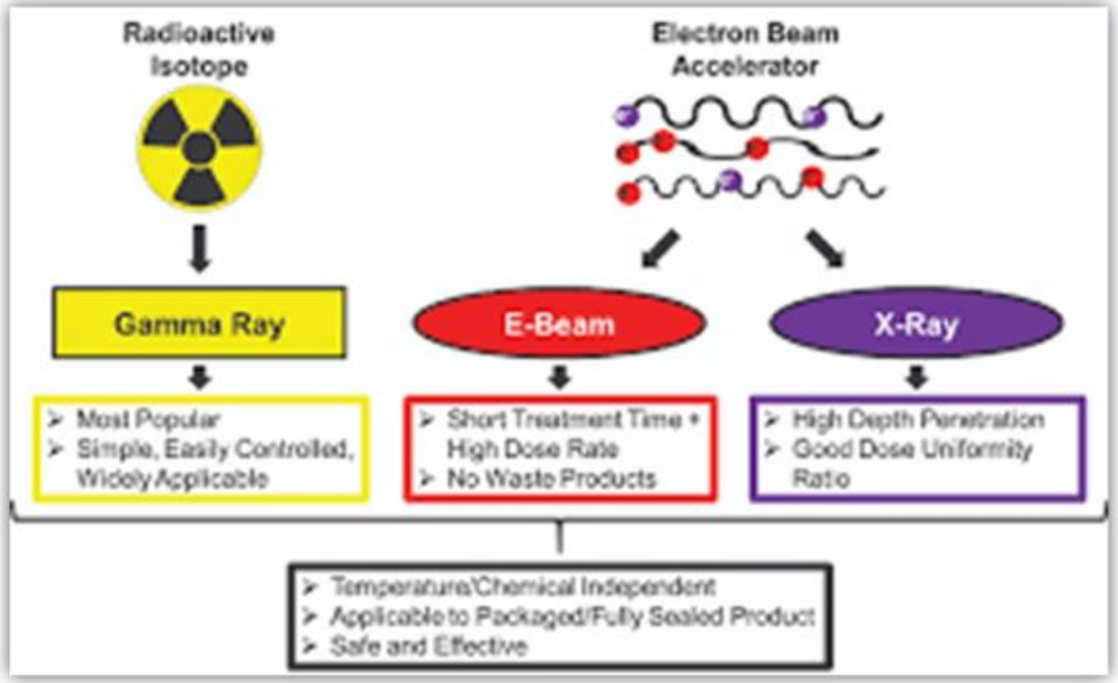
IRRADIATION CAN SERVE MANY PURPOSES

- **Delay of Sprouting and Ripening** – to **inhibit sprouting** (e.g., potatoes) and delay ripening of fruit to increase longevity.
- **Sterilization** – irradiation can be used to **sterilize foods**, which can then be **stored for years without refrigeration**. Sterilized foods are useful in hospitals for patients with severely impaired immune systems, such as patients with AIDS or undergoing chemotherapy.
- NASA astronauts eat meat that has been sterilized by irradiation to avoid getting foodborne illnesses when in space.

HOW IS FOOD IRRADIATED?

THERE ARE THREE SOURCES OF RADIATION APPROVED FOR USE ON FOODS

- Gamma rays are emitted from radioactive Cobalt 60 or Cesium 137. Gamma radiation is used routinely to sterilize medical, dental, and household products and is also used for the radiation treatment of cancer.
- X-rays can also be used to irradiate the food.
- Electron beam (or e-beam) is similar to X-rays and is a stream of high-energy electrons propelled from an electron accelerator into food.



HOW IS FOOD IRRADIATED?

DOES IRRADIATION ADVERSELY AFFECT THE NUTRITIONAL VALUE OF FOOD?

- The answer is No. In comparison to other food processing and preservation methods the nutritional value is least affected by irradiation.
- Extensive scientific studies have shown that irradiation has very little effect on the main nutrients such as proteins, carbohydrates, fats and minerals.

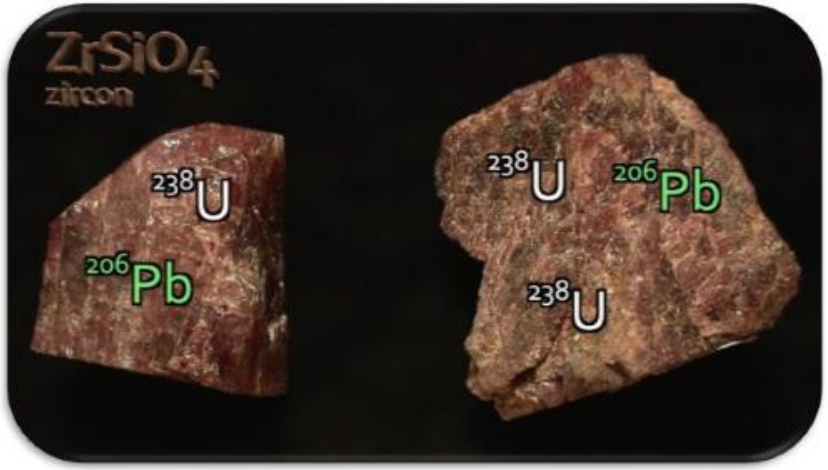
APPLICATIONS OF NUCLEAR TECHNOLOGY

2 PLANT MUTATION BREEDING

- Plant mutation breeding is the process of **exposing the seeds or cuttings of a given plant to radiation**, such as gamma rays, to cause mutations.
- The irradiated material is then cultivated to generate a sapling. Saplings are selected and multiplied if they show desired traits

3 CARBON DATING

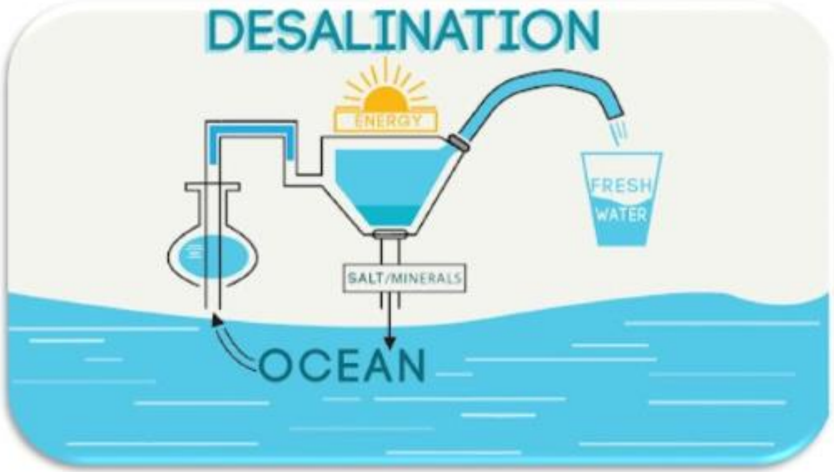
Analysing the relative abundance of particular naturally-occurring radioisotopes is of **vital importance in determining the age of rocks** and other materials that are of interest to geologists, anthropologists, hydrologists, and archaeologists, among others.



APPLICATIONS OF NUCLEAR TECHNOLOGY

4 DESALINATION

- **Potable water is a major priority in sustainable development.** Where it cannot be obtained from streams and aquifers, desalination of seawater, groundwater or urban waste water is required.
- **Most desalination today uses fossil fuels** and thus contributes to increased levels of greenhouse gases. **The feasibility of integrated nuclear desalination plants has been proven with experience in Kazakhstan, India, and Japan.**
- Large-scale deployment of nuclear desalination on a commercial basis with reactors built primarily for that purpose might be a reality very soon.



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5 MEDICINE

A DIAGNOSIS

- Diagnostic techniques in nuclear medicine use radiotracers which emit gamma rays from within the body. These tracers are generally short-lived isotopes linked to chemical compounds which permit specific physiological processes to be scrutinised.
- Dependent on the type of examination, radiotracers are either injected into the body, swallowed, or inhaled in gaseous form. The emissions from the radiotracers are detected by the imaging device, which provides pictures and molecular information.
- The superimposition of nuclear medicine images with computed tomography (CT) or magnetic resonance imaging (MRI) scans can provide comprehensive views to physicians to aid diagnosis.

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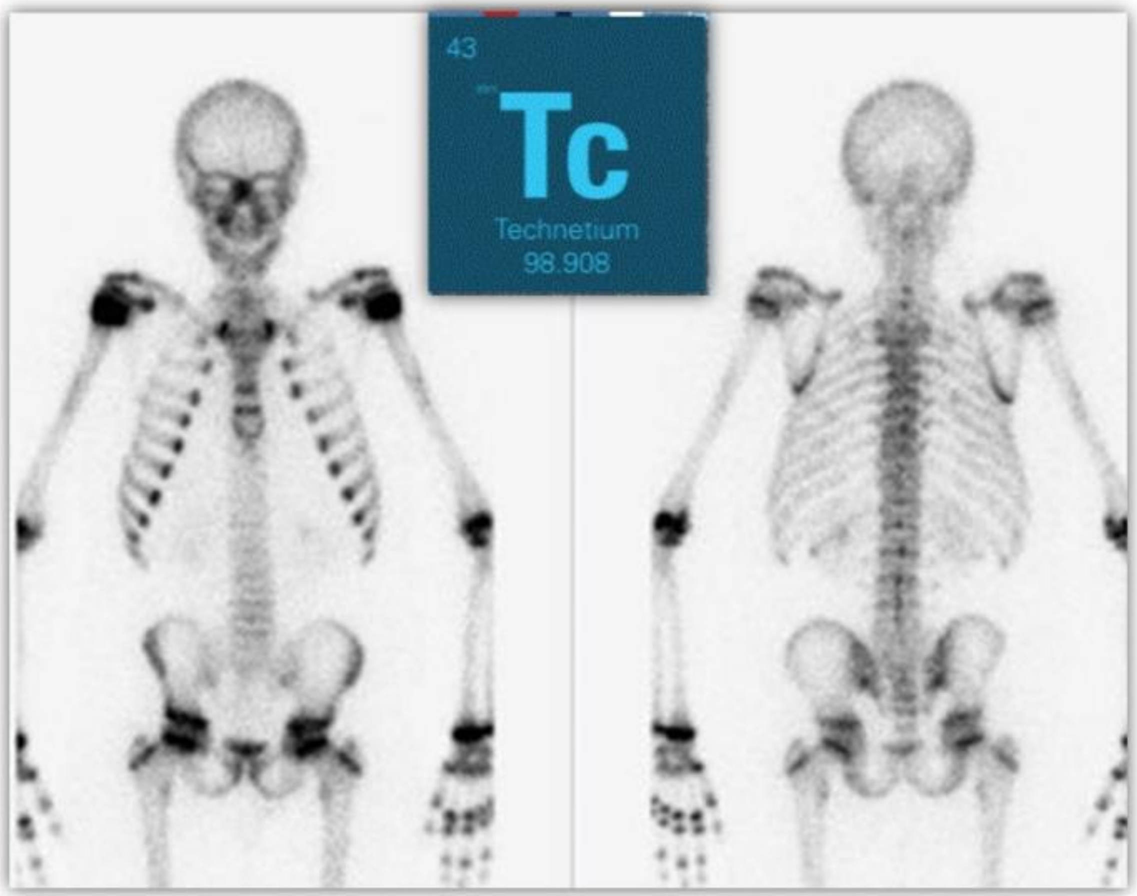
APPLICATIONS OF NUCLEAR TECHNOLOGY

5 MEDICINE

A DIAGNOSIS

- An advantage of nuclear over X-ray techniques is that both bone and soft tissue can be imaged very successfully.
- The most widely used diagnostic radioisotope is Technetium-99, with a half-life of six hours, and which gives the patient a very low radiation dose. Such isotopes are ideal for tracing many bodily processes with minimum discomfort to the patient. They are widely used to indicate tumours and to study the heart, lungs, liver, kidneys, blood circulation and bone structure.



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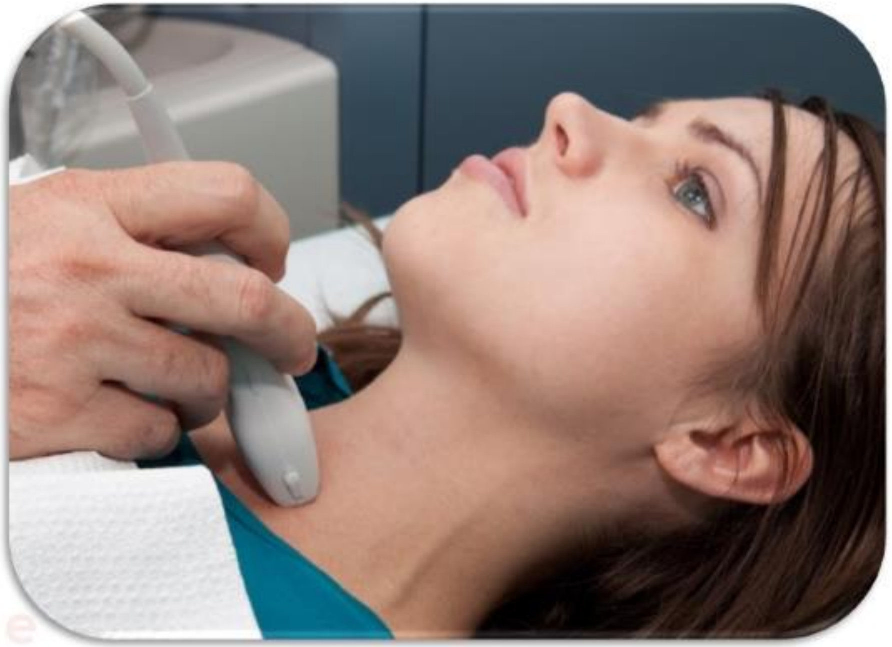
5 MEDICINE

B THERAPY

- Nuclear medicine is also used for therapeutic purposes. Most commonly, radioactive iodine (I-131) is used in small amounts to treat cancer and other conditions affecting the thyroid gland.
- The uses of radioisotopes in therapy are comparatively few, but important. Cancerous growths are sensitive to damage by radiation, which may be external (using a gamma beam from a cobalt-60 source), or internal (using a small gamma or beta radiation source).
- Short-range radiotherapy is known as brachytherapy, and this is becoming the main means of treatment. Many therapeutic procedures are palliative, usually to relieve pain.



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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